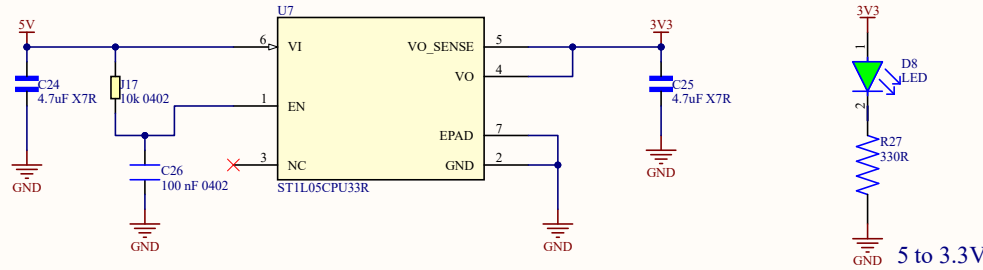


Add the out net from the toggle switch here , to choose in between the USB power and the power from the Buck 24 to 5 or 12 to 5



Splitting the 12V line is only half the battle; you must also manage the ground return paths.

When those three motors pull 6A, that 6A has to flow back to the battery through the ground plane. If that 6A flows underneath or through the ground copper near your STM32 or your TPS56637 feedback resistors, it will raise the local ground voltage, causing your microcontroller to read false button presses, corrupt SPI data, or trigger a brown-out reset.

How to route it in Altium:

Do not use one massive, uniform ground pour for the whole board.

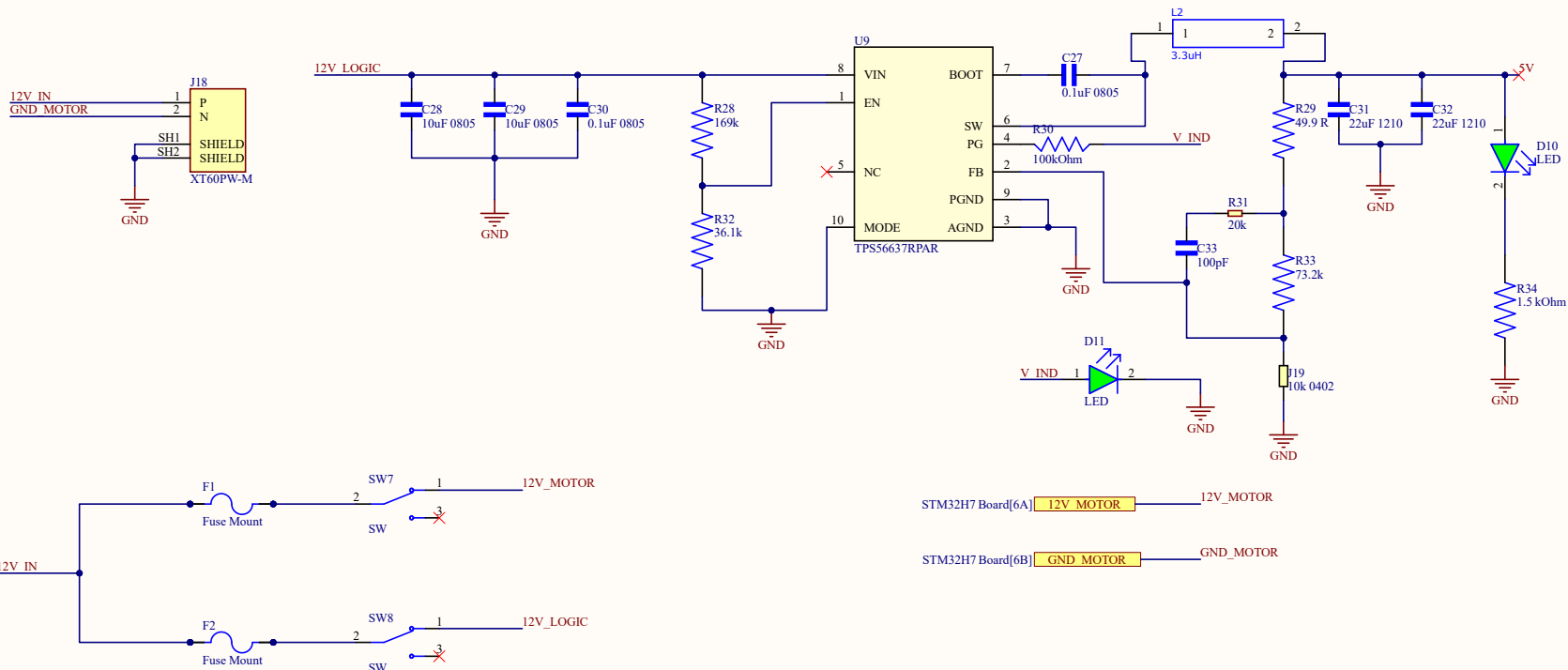
Create a specific, thick ground trace (or dedicated polygon pour) that connects the GND pins of the three TMC2209s directly back to the PGND pin of the XT60 connector.

Keep this motor ground return path physically away from your STM32 and analog sensors.

Your logic ground (PGND and AGND from the buck converter, and the STM32 ground) should meet the motor ground at exactly one single point: the physical solder pad of the XT60's negative pin.

3. Don't Forget the Bulk Capacitors!

As a final reminder, because you are running 3 motors from this 12V line, you absolutely must place a large aluminum electrolytic capacitor (e.g., 220uF to 470uF, rated for 35V or 50V) on the 12V_MOTORS net right next to the TMC2209 drivers. Without it, the inductive flyback from three simultaneous 2A motors will easily spike the 12V rail past 30V and detonate your chips.

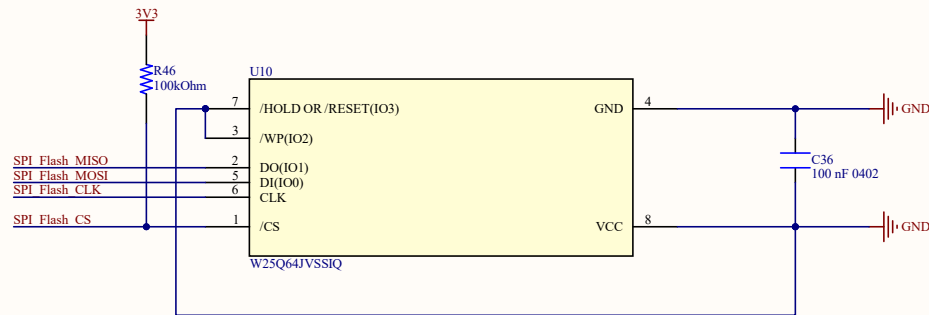
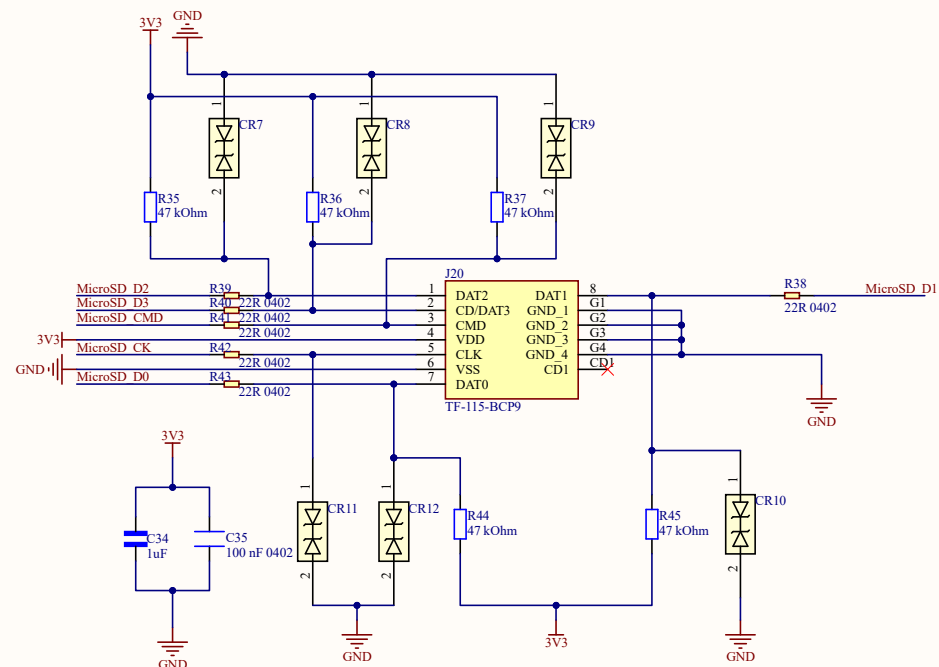


Title POWER		
Size B	Number 02	Revision 04
Date: 6/19/2026	Sheet of Neural Nexus	
File: C:\Work\Altium\EDR\POWER.SchDoc	Drawn By: Dulana Pitiwaduge	

Matched Net Lengths [Tol = 100 mil]

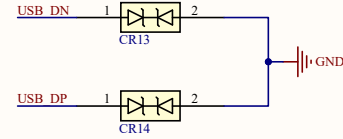
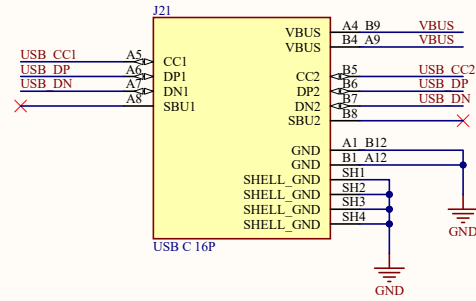
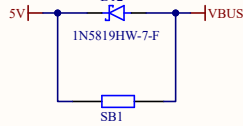
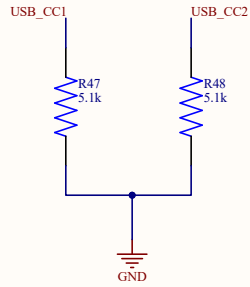
STM32H7 Board[1A]	MicroSD D3	MicroSD D3
STM32H7 Board[1A]	MicroSD D2	MicroSD D2
STM32H7 Board[1A]	MicroSD D1	MicroSD D1
STM32H7 Board[1A]	MicroSD D0	MicroSD D0
STM32H7 Board[1A]	MicroSD CK	MicroSD CK
STM32H7 Board[1A]	MicroSD CMD	MicroSD CMD

STM32H7 Board[1A]	SPI MISO	SPI_Flash_MISO
STM32H7 Board[1A]	SPI MOSI	SPI_Flash_MOSI
STM32H7 Board[1A]	SPI SCK	SPI_Flash_CLK
STM32H7 Board[1A]	SPI CS	SPI_Flash_CS

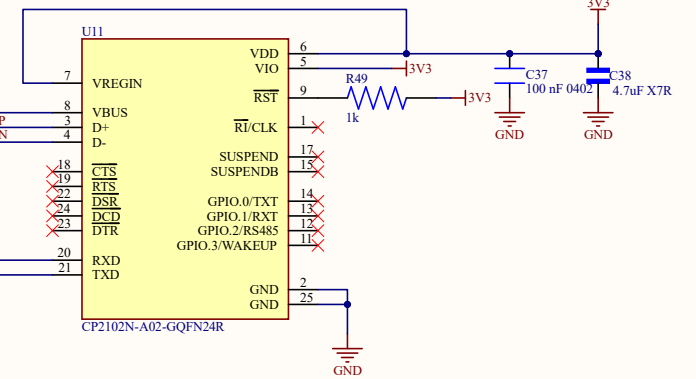
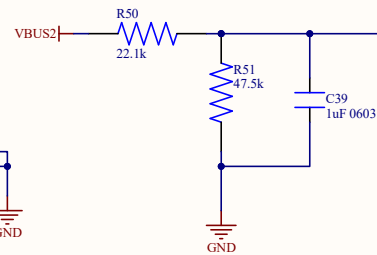
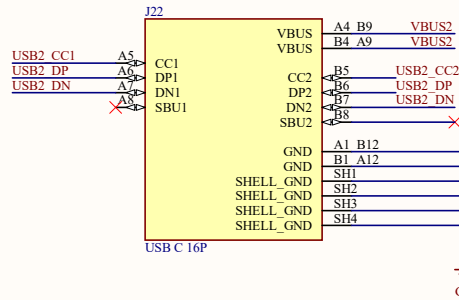
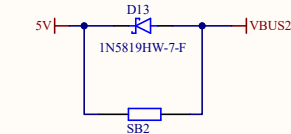
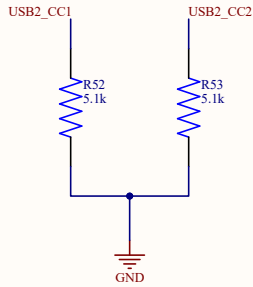


Title MicroSD & SPI FLASH		
Size B	Number 03	Revision 04
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STM32H7 Board[1A] USB1_DN
STM32H7 Board[1A] USB1_DP



STM32H7 Board[1A] USB2_TX
STM32H7 Board[1A] USB2_RX



Title PI_CON		
Size B	Number 04	Revision 01
Date: 6/19/2026	Sheet of Neural Nexus	
File: C:\Work\Altium\EDR\PI_CON.SchDoc	Drawn By: Dulana Pitiwaduge	